E-Commerce Sales & Profitability Analytics

Power BI Dashboard Project Description

Project Overview

This project focuses on analyzing e-commerce sales and profitability using an interactive Power BI dashboard. The dashboard turns sales data into clear visual insights about product categories, customer segments, shipping methods, geographic performance, and profit outcomes. It is designed to help users explore business performance and make more informed, data-driven decisions.

Objectives

- Monitor key performance indicators, including total sales, total profit, transaction count, and profit margin.
- Compare sales and profit across regions, product categories, and product sub-categories.
- Examine sales contributions by customer segment and shipping mode.
- Identify top-performing states and distinguish profitable, break-even, and loss-making records.
- Enable interactive exploration using slicers for Region, Category, Segment, Ship Mode, and City.

Tools & Technologies

Power BI Desktop for dashboard development; **DAX** for KPI measures and profit-status classification; and **Microsoft Excel** for supporting data review and preparation.

Dashboard Components

- KPI cards: Total Sales, Total Profit, Total Transactions, and Profit Margin.
- Charts: sales by category, regional sales and profit, profit status, segment contribution, shipping-mode contribution, top 10 states by sales, and sales/profit by sub-category.
- Interactive slicers: Region, Category, Segment, Ship Mode, and City.

Expected Business Value

The dashboard provides a consolidated view of sales and profitability, helps users compare performance across business dimensions, and highlights areas that may need further investigation, such as loss-making records or differences between sales and profit. Its interactive filters support focused analysis without requiring users to inspect the raw dataset manually.

Project Outcome

An interactive Power BI dashboard demonstrating practical skills in KPI development, DAX, data visualization, comparative analysis, and business-focused reporting.

Note: Numerical results should be reported only after verifying the final Power BI dashboard and the exact dataset version used.